

QUESTION -

2. Count Distinct Subsequences

Given a string **str**, Find the number of **distinct subsequences** that can be formed from it.

A subsequence is a sequence derived from the original string by deleting zero or more characters without changing the relative order of the remaining characters.

Note: Answer can be very large, so, output will be answer modulo **109+7**.

Examples:

Input: *str = "gfg"*

Output: 7

Explanation: The seven distinct subsequences are "", "g", "f", "gf", "fg", "gg" and "gfg"

Test Case	Input String	Expected Output
1	gfg	7
2	ggg	4
3	g	2
4	gg	3
5	gf	4

SOLUTION -

```
class Solution:
    def distinctSubsequences(self, s):
        MOD = 10**9 + 7

        # dp stores the number of distinct subsequences so far
        dp = 1

        # Stores dp value before the last occurrence of each character
        last = {}

        for ch in s:

            # Save current dp before updating
            previous_dp = dp

            # Every subsequence can either include or exclude current
            # character
            dp = (dp * 2) % MOD

            # If character appeared before,
            # subtract duplicate subsequences
            if ch in last:
                dp = (dp - last[ch] + MOD) % MOD

            # Store previous dp for future duplicates
            last[ch] = previous_dp

        return dp
```